Influence of Social Media Recommendation Algorithms on User Behavior

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***Abstract*—**Social media recommendation systems increasingly determine which posts, videos, accounts, and discussions users encounter. Although these systems are designed to improve relevance and reduce information overload, their ranking decisions can also shape attention, engagement, content preferences, and patterns of repeated use. This paper examines how recommendation algorithms influence user behavior through personalization, engagement optimization, feedback loops, and selective exposure. A literature-based analytical approach is used to synthesize findings from research on social recommender systems, short-form video platforms, algorithmic curation, and recent field experiments. The reviewed evidence indicates that user behavior and algorithmic ranking form a reciprocal process: past interactions affect future recommendations, while recommended content changes the opportunities available for subsequent interaction. This process can increase convenience and discovery, but it may also reduce content diversity, reinforce existing interests, and create persistent engagement patterns. Recent experimental evidence further shows that algorithmic feeds can influence political attitudes and following behavior, demonstrating that recommendation effects may extend beyond simple clicking or viewing. The paper proposes a research framework for measuring algorithmic influence through exposure, engagement, persistence, diversity, and perceived user control. It concludes that recommendation quality should be evaluated not only by engagement metrics but also by user well-being, transparency, content diversity, and long-term behavioral outcomes.

***Keywords*—**social media, recommendation systems, algorithmic curation, user behavior, personalization, engagement, filter bubbles, short-form video

I. I. INTRODUCTION

Social media has changed from a collection of user-selected pages into an algorithmically organized information environment. On many platforms, users no longer receive content solely because they explicitly follow an account. Instead, ranking systems select and order material according to signals inferred from previous behavior, social connections, content characteristics, and contextual information. Recommendation systems therefore act as an intermediate layer between the large volume of available content and the small amount of content that a person actually sees. Research on social recommender systems has shown that personalization can reduce information overload while improving the relevance of suggestions [1], [2].

The same mechanism creates an important research problem. If the system learns from clicks, viewing duration, likes, shares, follows, and other interactions, those interactions can influence the next set of recommendations. The user is therefore not simply receiving recommendations; the user is participating in a feedback loop that continuously updates the information environment. Studies of TikTok, for example, found that follow actions, likes, and viewing behavior can materially change subsequent recommendations [3]. Large-scale data-donation research has also shown how recommendation-driven short-video consumption can be examined through actual viewing histories rather than self-reported usage alone [4].

The influence of recommendation algorithms should consequently be evaluated at the behavioral level. A recommendation can affect whether a person clicks, how long the person remains on a platform, whether similar content is consumed repeatedly, and whether new accounts or topics are followed. At a broader level, algorithmic curation may influence the diversity of information a user encounters and may contribute to political or social attitude changes. Recent experimental evidence from X indicates that switching users to an algorithmic feed increased engagement and shifted several political attitudes, with some behavioral changes persisting after the algorithmic treatment ended [5].

This paper analyzes these mechanisms through a literature-based research approach. The objectives are to identify the principal pathways through which recommendation algorithms affect user behavior, distinguish beneficial personalization from potentially harmful reinforcement, and propose measurable variables for future empirical studies. The paper does not claim that algorithms alone determine user behavior. Instead, it treats behavior as the outcome of an interaction between algorithmic ranking, individual preferences, social context, and platform incentives.

II. II. BACKGROUND AND RELATED WORK

Recommender systems were originally developed to help users navigate large collections of items. In social networks, however, the recommendation problem is more complex because the available items are continuously produced by users, interactions are socially connected, and preferences can change rapidly. Campana and Delmastro describe social recommender systems as systems that combine conventional recommendation techniques with information about users, relationships, content, and social context [1]. Machine-learning approaches include collaborative filtering, content-based methods, network-based recommendation, and hybrid approaches [1], [2].

Personalization is not inherently harmful. It can make information retrieval faster, expose users to relevant material, and help smaller creators reach interested audiences. A scoping review of personalized social-media experiences identified user awareness, user control, algorithmic and social curation, and echo-chamber effects as important themes in the literature [6]. This suggests that the central research question is not whether personalization exists, but how its objectives and feedback mechanisms shape the resulting user experience.

Research on short-form video platforms provides a useful example because the interaction cycle is rapid. Boeker and Urman found that language, location, follows, likes, and viewing behavior affected TikTok recommendations, with following behavior having particularly strong influence in their audit [3]. Zannettou and colleagues later analyzed 9.2 million recommended videos from 347 TikTok users and examined time spent, attention, and interaction as indicators of engagement [4]. These studies demonstrate why recommendation effects should be studied using both algorithmic exposure and actual user response.

A further line of research concerns the relationship between engagement optimization and user welfare. Khambatta and colleagues experimentally compared recommendations based on actual preferences with recommendations designed around ideal preferences. Their results showed that actual-preference recommendations generated more clicks, while ideal-preference recommendations produced several measures of improved perceived value and time well spent [7]. This finding challenges the assumption that the recommendation with the highest immediate engagement is necessarily the best recommendation for the user.

III. III. RECOMMENDATION ALGORITHM AND BEHAVIORAL FEEDBACK

A recommendation system can be understood as a repeated prediction-and-response process. At time t, the platform observes a user profile and recent interaction history, generates candidate content, ranks candidates, and displays a selected set. The user's response then becomes new input for later recommendations. A simplified behavioral loop is: User history → candidate generation → ranking → exposure → interaction → updated user representation → new ranking.

The loop is important because exposure is not neutral. Content that receives more opportunities to be seen has more opportunities to be clicked, liked, shared, or discussed. A highly engaged item can therefore gain additional visibility, producing more interactions and reinforcing its ranking. Recent theoretical work models this as a dynamic feedback loop in which engagement-oriented ranking can increase platform engagement while also increasing the risk of misinformation and polarization [8].

At the individual level, the loop can produce reinforcement of interests. A user who repeatedly watches a topic may receive more of that topic, which reduces the relative visibility of unrelated topics. The resulting behavior can look like a strong personal preference even when the original preference was weak. This distinction matters because the system learns from behavior that it has partly helped create.

However, the feedback loop should not be interpreted as proof of manipulation in every case. Users make choices, actively search for information, follow accounts, skip content, and sometimes reject recommendations. Evidence from Google Search, for example, indicates that users' own choices can account for substantial differences between what they are exposed to and what they ultimately engage with [9]. Therefore, algorithmic influence and user agency should be treated as interacting variables rather than competing explanations.

IV. IV. MAJOR AREAS OF INFLUENCE

A. *A. Attention and Time Spent*

Recommendation systems can reduce the effort required to find the next item, especially when content is presented as an uninterrupted feed. This lowers the cost of continuing from one item to another. Data-donation research on TikTok found that average daily usage time increased over users' observed platform lifetime, while attention to individual videos remained relatively stable [4]. The finding does not establish that recommendations alone caused the increase, but it demonstrates the value of measuring time and attention separately. A user may remain on a platform longer without becoming proportionally more attentive to each item.

B. *B. Engagement and Interaction*

Engagement is commonly represented by measurable actions such as clicks, likes, comments, shares, follows, saves, or viewing completion. These actions can become ranking signals, which creates a direct connection between user behavior and future exposure. Research comparing actual and ideal preference recommendations found higher click-through for actual-preference recommendations, while ideal-preference recommendations produced better outcomes on several user-centered measures [7]. This indicates that maximizing a narrow engagement metric can produce a different outcome from maximizing perceived user benefit.

C. *C. Content Diversity and Selective Exposure*

Personalization can improve relevance while reducing the probability that users encounter unrelated information. A personalized environment may therefore become narrower over time. Research on Facebook has documented substantial exposure to like-minded sources, while also showing that such exposure did not automatically translate into increased polarization in the studied setting [10]. This is an important caution against treating filter bubbles as a universal or deterministic effect. Diversity, exposure, and attitudinal change should be measured separately.

D. *D. Opinion and Attitude Formation*

Recommendation effects can extend beyond entertainment behavior. The 2026 field experiment on X found that exposure to the platform's algorithmic feed increased engagement and shifted several political attitudes, while also changing account-following behavior [5]. The result is significant because following an account is a persistent behavioral action that can influence future information exposure. It suggests that algorithmic effects may have a longer time horizon than the immediate interaction used to rank content.

E. *E. Misinformation and Polarization*

Engagement-oriented ranking can create incentives for content that attracts strong reactions. A 2026 study by Germano, Gómez, and Sobbrio models how popularity-based ranking interacts with user behavior and finds a trade-off between engagement and risks such as misinformation and polarization [8]. This does not mean that every highly engaging item is harmful. Rather, it suggests that the objective function used by a recommender system matters: optimizing only for interaction can produce different information outcomes from optimizing for quality, diversity, or long-term user welfare.

F. *F. User Awareness and Control*

Users are not always able to determine why a particular recommendation appears. Limited understanding can affect trust, perceived autonomy, and the ability to correct unwanted personalization. The social-media personalization literature identifies algorithmic awareness and user control as recurring concerns [6]. Greater transparency alone may not solve the problem, but understandable explanations, feed controls, preference reset options, and meaningful feedback mechanisms can give users more influence over the information environment.

V. V. SYNTHESIS OF RESEARCH EVIDENCE

Study focus	Observed relationship	Research implication
Social recommender systems [1], [2]	Social and behavioral data improve personalization	Recommendation is context-dependent, not only content-dependent
TikTok personalization [3]	Follows, likes, viewing and context affect recommendations	Small interactions can alter later exposure
TikTok data donation [4]	Longer usage and measurable attention patterns	Behavior should be studied using real interaction traces
Ideal versus actual preferences [7]	Engagement and perceived user benefit can diverge	Clicks alone are insufficient as a quality metric
X field experiment [5]	Algorithmic feed affected engagement and political attitudes	Effects may persist beyond immediate exposure
Engagement ranking [8]	Feedback can amplify misinformation and polarization	Objective functions require multi-dimensional evaluation

Table I summarizes the principal evidence used in this paper. The studies differ in platform, methodology, country, and outcome, so their results should not be treated as directly interchangeable. Taken together, however, they support a common interpretation: recommendation algorithms influence the opportunity structure in which user choices occur, while users simultaneously provide the behavioral signals that shape later recommendations.

VI. VI. PROPOSED RESEARCH FRAMEWORK

To study the influence of recommendation algorithms in a controlled academic project, this paper proposes a mixed-method framework combining user surveys with behavioral measurements. The framework is designed to avoid the common mistake of treating screen time as the only outcome. Instead, influence is represented through five dimensions: exposure, engagement, persistence, diversity, and perceived control.

Exposure measures the proportion of viewed items that were recommended rather than explicitly searched for or selected from followed accounts. Engagement measures actions such as watch completion, likes, comments, shares, and saves. Persistence measures whether the user continues interacting with the same topic or creator after repeated recommendations. Diversity measures the range of topics, sources, or viewpoints encountered. Perceived control measures whether users understand and feel able to modify the recommendation process.

A future survey can collect demographic information, platform usage frequency, perceived algorithmic awareness, and self-reported changes in behavior. Where ethically and technically feasible, participants can optionally provide exported usage data. The study should avoid collecting private messages, passwords, or unnecessary personal identifiers. Survey responses can then be compared with behavioral measures using correlation, chi-square tests, or regression analysis depending on the research questions and variable types.

A. *A. Research Questions*

RQ1: To what extent does exposure to recommended content relate to users' engagement and time spent on social media?

RQ2: Does repeated recommendation of a topic predict continued interaction with similar content?

RQ3: Is higher personalization associated with lower perceived content diversity?

RQ4: Does algorithmic awareness change the relationship between recommendations and user behavior?

RQ5: Do users consider engagement-oriented recommendations beneficial to their long-term interests?

B. *B. Variables*

Independent variables may include recommendation frequency, perceived personalization, recommendation source, and algorithmic awareness. Dependent variables may include daily usage duration, interaction frequency, topic persistence, content diversity, and perceived satisfaction. Control variables can include age group, primary platform, baseline usage, and frequency of active searching. The framework allows a future empirical study to distinguish association from stronger causal claims.

VII. VII. ETHICAL AND DESIGN CONSIDERATIONS

Recommendation research involves a direct ethical issue because the object of study is a system that can change what participants see. A study should therefore avoid deliberately exposing participants to harmful or deceptive material merely to create an experimental effect. Informed consent, data minimization, secure storage, anonymization, and the right to withdraw should be part of the research design.

Platform designers also face a multi-objective optimization problem. A system can be accurate in predicting clicks while performing poorly on diversity, user well-being, or misinformation exposure. The evidence reviewed here suggests that a more responsible objective should balance relevance and engagement with longer-term measures of user value [7], [8]. Transparency should similarly be treated as a functional feature rather than a decorative notice. Users should have understandable ways to inspect, reset, or influence important recommendation preferences.

VIII. VIII. DISCUSSION

The reviewed literature suggests that recommendation algorithms should be understood as behavioral environments rather than simple content-selection tools. Their influence emerges from repeated interaction. A recommendation changes what is available to notice; the resulting interaction becomes evidence for the next ranking decision; the updated ranking changes subsequent exposure. This recursive process can be beneficial when it helps users discover useful content, but it can become problematic when the system repeatedly rewards narrow or emotionally intense signals.

An important finding across the literature is that algorithmic influence is neither fully deterministic nor fully negligible. User choices matter, and some users deliberately search for particular content or follow accounts that match their interests [9]. At the same time, ranking systems decide which options receive greater visibility. The distinction between choice and exposure is therefore essential. A user cannot choose content that is rarely shown, while content that is repeatedly presented has more opportunities to generate interaction.

Another important implication is that engagement should not be treated as a universal proxy for satisfaction. The experiment by Khambatta and colleagues demonstrates that recommendations optimized for actual preferences can produce more clicks while recommendations aligned with ideal preferences can improve several user-centered outcomes [7]. This suggests that future recommendation evaluation should include measures such as time well spent, perceived usefulness, content diversity, and post-session satisfaction.

Finally, recent evidence on X indicates that recommendation effects can extend into durable behavioral outcomes, including account-following and political attitudes [5]. This raises the importance of longitudinal research. Short laboratory studies may capture immediate reactions but miss cumulative effects produced by months of repeated algorithmic exposure. Future studies should therefore combine short-term interaction metrics with longer-term measures of preference change, following behavior, and information diversity.

IX. IX. LIMITATIONS AND FUTURE WORK

This paper is a literature-based analytical study and does not present a new participant dataset. Consequently, it cannot estimate the causal effect of recommendation algorithms for a specific population. The reviewed studies also use different platforms, countries, time periods, and methodologies. Platform algorithms are proprietary and can change rapidly, which makes results difficult to generalize across services or across time.

Future research should address these limitations through longitudinal, multi-platform studies. A particularly useful design would compare algorithmic and chronological feeds while recording both exposure and engagement. Another promising direction is the development of recommendation-quality metrics that jointly evaluate relevance, diversity, user welfare, and harmful-content exposure. Data-donation methods can provide richer behavioral evidence while preserving participant control when privacy safeguards are properly implemented [4].

Researchers should also investigate differences among age groups, levels of algorithmic literacy, and types of social media use. Entertainment recommendations may influence attention differently from news recommendations, while recommendations involving friends may have different social consequences from recommendations involving unknown creators. These distinctions can produce more precise theories than the broad claim that algorithms simply influence behavior.

X. X. CONCLUSION

Social media recommendation algorithms have become a major mechanism through which users encounter digital content. Their influence arises from a continuous feedback relationship between algorithmic ranking and human behavior. Personalization can improve relevance, discovery, and convenience, but repeated optimization for engagement can also reinforce narrow interests, alter exposure patterns, and create risks related to misinformation, polarization, and reduced user autonomy.

The evidence reviewed in this paper supports a balanced interpretation. Algorithms matter, but they operate together with individual choice, social context, and platform incentives. Therefore, responsible recommendation research and design should move beyond the single question of whether a system increases engagement. It should also ask whether users encounter diverse and useful information, whether they feel in control, and whether the resulting behavior is beneficial over time. A multi-dimensional evaluation framework can help researchers and platforms pursue recommendation systems that are not only more engaging, but also more transparent, diverse, and aligned with long-term user interests.

XI. REFERENCES

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